EXPERIMENT 5

1 Update Ubuntu:

```
vboxuser@Mona:~/CloudSchedulingLab$ sudo apt update
[sudo: authenticate] Password:
Hit:1 http://security.ubuntu.com/ubuntu resolute-security InRelease
Hit:2 http://in.archive.ubuntu.com/ubuntu resolute InRelease
Hit:3 http://in.archive.ubuntu.com/ubuntu resolute-updates InRelease
Hit:4 http://in.archive.ubuntu.com/ubuntu resolute-backports InRelease
16 packages can be upgraded. Run 'apt list --upgradable' to see them.
```

```
vboxuser@Mona:~/CloudSchedulingLab$ sudo apt upgrade -y
The following packages were automatically installed and are no longer required:
  linux-headers-7.0.0-31          linux-image-7.0.0-31-generic            linux-modules-7.0.0-31-generic  linux-tools-7.0.0-31-generic
  linux-headers-7.0.0-31-generic  linux-main-modules-zfs-7.0.0-31-generic  linux-tools-7.0.0-31
Use 'sudo apt autoremove' to remove them.

Get more security updates through Ubuntu Pro with 'esm-apps' enabled:
  python3-pip-whl python3-pip
Learn more about Ubuntu Pro at https://ubuntu.com/pro
Upgrading:
  gir1.2-soup-3.0   libfreerdp-server3-3   libheif-plugin-aomenc  libsoup-3.0-0       openjdk-17-jdk           openjdk-17-jre-headless
  heif-gdk-pixbuf   libfreerdp3-3         libheif1               libsoup-3.0-common  openjdk-17-jdk-headless
  heif-thumbnailer  libheif-plugin-aomdec  libnfs14               libwinpr3-3         openjdk-17-jre

Summary:
  Upgrading: 16, Installing: 0, Removing: 0, Not Upgrading: 0
16 standard LTS security updates
  Download size: 124 MB
  Space needed: 476 kB / 41.1 GB available

Get:1 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 libsoup-3.0-common all 3.6.6-1ubuntu0.3 [9,690 B]
Get:2 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 gir1.2-soup-3.0 amd64 3.6.6-1ubuntu0.3 [22.5 kB]
Get:3 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 libsoup-3.0-0 amd64 3.6.6-1ubuntu0.3 [295 kB]
Get:4 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 libheif-plugin-aomenc amd64 1.21.2-3ubuntu0.7 [20.9 kB]
Get:5 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 heif-gdk-pixbuf amd64 1.21.2-3ubuntu0.7 [7,762 B]
Get:6 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 heif-thumbnailer amd64 1.21.2-3ubuntu0.7 [14.8 kB]
Get:7 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 libheif1 amd64 1.21.2-3ubuntu0.7 [551 kB]
Get:8 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 libheif-plugin-aomdec amd64 1.21.2-3ubuntu0.7 [15.6 kB]
Get:9 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 libfreerdp-server3-3 amd64 3.32.2+dfsg-0ubuntu0.26.04.1 [103 kB]
Get:10 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 libfreerdp3-3 amd64 3.32.2+dfsg-0ubuntu0.26.04.1 [829 kB]
Get:11 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 libwinpr3-3 amd64 3.32.2+dfsg-0ubuntu0.26.04.1 [411 kB]
Get:12 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 libnfs14 amd64 5.0.2-1ubuntu1.2 [110 kB]
```

2 Install Java, Maven and Git:

```
vboxuser@Mona:~/CloudSchedulingLab$ sudo apt install openjdk-17-jdk maven git -y
openjdk-17-jdk is already the newest version (17.0.21.0.3+1-1~26.04.1).
maven is already the newest version (3.9.12-1).
git is already the newest version (1:2.53.0-1ubuntu1.1).
The following packages were automatically installed and are no longer required:
  linux-headers-7.0.0-31          linux-image-7.0.0-31-generic            linux-modules-7.0.0-31-generic  linux-tools-7.0.0-31-generic
  linux-headers-7.0.0-31-generic  linux-main-modules-zfs-7.0.0-31-generic  linux-tools-7.0.0-31
Use 'sudo apt autoremove' to remove them.

Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 0
vboxuser@Mona:~/CloudSchedulingLab$ java -version
openjdk version "17.0.21.0.3" 2026-10-24
OpenJDK Runtime Environment (build 17.0.21.0.3+1-1-26.04.1-Ubuntu)
OpenJDK 64-Bit Server VM (build 17.0.21.0.3+1-1-26.04.1-Ubuntu, mixed mode, sharing)
vboxuser@Mona:~/CloudSchedulingLab$ javac -version
javac 17.0.21.0.3
vboxuser@Mona:~/CloudSchedulingLab$ mvn -version
Apache Maven 3.9.12
Maven home: /usr/share/maven
Java version: 17.0.21.0.3, vendor: Ubuntu, runtime: /usr/lib/jvm/java-17-openjdk-amd64
Default locale: en_US, platform encoding: UTF-8
OS name: "linux", version: "7.0.0-43-generic", arch: "amd64", family: "unix"
vboxuser@Mona:~/CloudSchedulingLab$ git --version
git version 2.53.0
```

3. Create the CloudSim Plus Maven project:

3.1 Create folders:

```
vboxuser@Mona:~/CloudSchedulingLab$ mkdir -p ~/CloudSchedulingLab/src/main/java/lab
vboxuser@Mona:~/CloudSchedulingLab$ cd ~/CloudSchedulingLab
```

3.2 Create pom.xml:

```
vboxuser@Mona:~/CloudSchedulingLab$ cd ~/CloudSchedulingLab
vboxuser@Mona:~/CloudSchedulingLab$ nano pom.xml
```

```
 GNU nano 8.7.1                                  pom.xml
<project xmlns="http://maven.apache.org/POM/4.0.0"
    xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
https://maven.apache.org/xsd/maven-4.0.0.xsd">
<modelVersion>4.0.0</modelVersion>
<groupId>lab</groupId>
<artifactId>cloud-scheduling-lab</artifactId>
<version>1.0-SNAPSHOT</version>
<properties>
<maven.compiler.release>17</maven.compiler.release>
<project.build.sourceEncoding>UTF-8</project.build.sourceEncoding>
</properties>
<dependencies>
<dependency>
<groupId>org.cloudsimplus</groupId>
<artifactId>cloudsimplus</artifactId>
<version>8.5.5</version>
</dependency>
</dependencies>
</project>
```

3.3 Resolve dependencies:

```
vboxuser@Mona:~/CloudSchedulingLab$ mvn dependency:resolve
[INFO] Scanning for projects...
[INFO]
[INFO] ----------------------< lab:cloud-scheduling-lab >----------------------
[INFO] Building cloud-scheduling-lab 1.0-SNAPSHOT
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- dependency:3.7.0:resolve (default-cli) @ cloud-scheduling-lab ---
[INFO]
[INFO] The following files have been resolved:
[INFO]    org.cloudsimplus:cloudsimplus:jar:8.5.5:compile -- module cloudsimplus (auto)
[INFO]    org.apache.commons:commons-math3:jar:3.6.1:compile -- module commons.math3 (auto)
[INFO]    org.apache.commons:commons-lang3:jar:3.12.0:compile -- module org.apache.commons.lang3 [auto]
[INFO]    org.apache.commons:commons-collections4:jar:4.4:compile -- module org.apache.commons.collections4 [auto]
[INFO]    com.google.code.gson:gson:jar:2.10.1:compile -- module com.google.gson
[INFO]    ch.qos.logback:logback-core:jar:1.4.12:compile -- module ch.qos.logback.core
[INFO]    ch.qos.logback:logback-classic:jar:1.4.6:compile -- module ch.qos.logback.classic
[INFO]    org.slf4j:slf4j-api:jar:2.0.4:compile -- module org.slf4j
[INFO]
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  1.045 s
[INFO] Finished at: 2026-11-10T04:04:46Z
[INFO] ------------------------------------------------------------------------
vboxuser@Mona:~/CloudSchedulingLab$ mvn clean
[INFO] Scanning for projects...
[INFO]
[INFO] ----------------------< lab:cloud-scheduling-lab >----------------------
[INFO] Building cloud-scheduling-lab 1.0-SNAPSHOT
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- clean:3.2.0:clean (default-clean) @ cloud-scheduling-lab ---
[INFO] Deleting /home/vboxuser/CloudSchedulingLab/target
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
```

4. Implement the experiment:

```
[INFO] Finished at: 2026-11-10T04:04:58Z
[INFO] ------------------------------------------------------------------------
vboxuser@Mona:~/CloudSchedulingLab$ nano src/main/java/lab/SchedulingExperiment.java
```

```
 GNU nano 8.7.1                          src/main/java/lab/SchedulingExperiment.java
package lab;

import java.util.ArrayList;
import java.util.List;

import org.cloudsimplus.core.CloudSimPlus;
import org.cloudsimplus.datacenters.Datacenter;
import org.cloudsimplus.datacenters.DatacenterSimple;
import org.cloudsimplus.brokers.DatacenterBroker;
import org.cloudsimplus.brokers.DatacenterBrokerSimple;
import org.cloudsimplus.cloudlets.Cloudlet;
import org.cloudsimplus.cloudlets.CloudletSimple;
import org.cloudsimplus.hosts.Host;
import org.cloudsimplus.hosts.HostSimple;
import org.cloudsimplus.resources.Pe;
import org.cloudsimplus.resources.PeSimple;
import org.cloudsimplus.schedulers.cloudlet.CloudletScheduler;
import org.cloudsimplus.schedulers.cloudlet.CloudletSchedulerSpaceShared;
import org.cloudsimplus.schedulers.cloudlet.CloudletSchedulerTimeShared;
import org.cloudsimplus.schedulers.vm.VmScheduler;
import org.cloudsimplus.schedulers.vm.VmSchedulerSpaceShared;
import org.cloudsimplus.schedulers.vm.VmSchedulerTimeShared;
import org.cloudsimplus.utilizationmodels.UtilizationModel;
import org.cloudsimplus.utilizationmodels.UtilizationModelFull;
import org.cloudsimplus.vms.Vm;
import org.cloudsimplus.vms.VmSimple;
import org.cloudsimplus.provisioners.ResourceProvisionerSimple;
import org.cloudsimplus.allocationpolicies.VmAllocationPolicySimple;

public class SchedulingExperiment {
    private static final int HOST_PES = 4;
    private static final int VM_COUNT = 2;
    private static final int CLOUDLET_COUNT = 4;
    private static final long MIPS = 1000;
    private static final long CLOUDLET_LENGTH = 20000;
```

```java
public class SchedulingExperiment {
    private static final int HOST_PES = 4;
    private static final int VM_COUNT = 2;
    private static final int CLOUDLET_COUNT = 4;
    private static final long MIPS = 1000;
    private static final long CLOUDLET_LENGTH = 20000;

    public static void main(String[] args) {
        runScenario("A: VM SpaceShared + Cloudlet SpaceShared",
                new VmSchedulerSpaceShared(),
                new CloudletSchedulerSpaceShared());
        runScenario("B: VM SpaceShared + Cloudlet TimeShared",
                new VmSchedulerSpaceShared(),
                new CloudletSchedulerTimeShared());
        runScenario("C: VM TimeShared + Cloudlet SpaceShared",
                new VmSchedulerTimeShared(),
                new CloudletSchedulerSpaceShared());
        runScenario("D: VM TimeShared + Cloudlet TimeShared",
                new VmSchedulerTimeShared(),
                new CloudletSchedulerTimeShared());
    }

    private static void runScenario(String title,
                                    VmScheduler vmScheduler,
                                    CloudletScheduler cloudletScheduler) {
        System.out.println("\n================================================");
        System.out.println(title);
        System.out.println("================================================");

        CloudSimPlus simulation = new CloudSimPlus();
        createDatacenter(simulation, vmScheduler);
        DatacenterBroker broker = new DatacenterBrokerSimple(simulation);
```

```java
                                CloudletScheduler cloudletScheduler) {
    System.out.println("\n================================================");
    System.out.println(title);
    System.out.println("================================================");

    CloudSimPlus simulation = new CloudSimPlus();
    createDatacenter(simulation, vmScheduler);
    DatacenterBroker broker = new DatacenterBrokerSimple(simulation);

    List<Vm> vmList = createVms(cloudletScheduler);
    List<Cloudlet> cloudletList = createCloudlets();

    broker.submitVmList(vmList);
    broker.submitCloudletList(cloudletList);
    simulation.start();

    List<Cloudlet> finished = broker.getCloudletFinishedList();
    System.out.printf("%-10s %-10s %-12s %-12s %-12s%n",
            "Cloudlet", "VM", "Status", "Start", "Finish");
    for (Cloudlet cloudlet : finished) {
        System.out.printf("%-10d %-10d %-12s %-12.2f %-12.2f%n",
                cloudlet.getId(), cloudlet.getVm().getId(),
                cloudlet.getStatus(), cloudlet.getStartTime(),
                cloudlet.getFinishTime());
    }
    System.out.println("Completed Cloudlets: " + finished.size()
            + " / " + CLOUDLET_COUNT);


rivate static Datacenter createDatacenter(CloudSimPlus simulation,
                                          VmScheduler vmScheduler) {
    List<Pe> peList = new ArrayList<>();
    for (int i = 0; i < HOST_PES; i++) {
```

```
        hostList.add(host);
        return new DatacenterSimple(simulation, hostList,
                new VmAllocationPolicySimple());
    }

    private static List<Vm> createVms(CloudletScheduler scheduler) {
        List<Vm> vmList = new ArrayList<>();
        for (int i = 0; i < VM_COUNT; i++) {
            CloudletScheduler newScheduler;
 (scheduler instanceof CloudletSchedulerTimeShared){
ewScheduler = new CloudletSchedulerTimeShared();
lse{
ewScheduler = new CloudletSchedulerSpaceShared();

            Vm vm = new VmSimple(i, MIPS, 2);
            vm.setRam(2048).setBw(10000).setSize(10000);
            vm.setCloudletScheduler(newScheduler);
            vmList.add(vm);
        }
        return vmList;
    }

    private static List<Cloudlet> createCloudlets() {
        List<Cloudlet> cloudletList = new ArrayList<>();
        UtilizationModel utilization = new UtilizationModelFull();
        for (int i = 0; i < CLOUDLET_COUNT; i++) {
            Cloudlet cloudlet = new CloudletSimple(
                    CLOUDLET_LENGTH, 1, utilization);
            cloudlet.setSizes(300);
            cloudletList.add(cloudlet);
        }
        return cloudletList;
    }
```

5. Compile and execute:

5.1 Compile:

```
vboxuser@Mona:~/CloudSchedulingLab$ mvn clean compile
[INFO] Scanning for projects...
[INFO]
[INFO] ----------------------< lab:cloud-scheduling-lab >----------------------
[INFO] Building cloud-scheduling-lab 1.0-SNAPSHOT
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- clean:3.2.0:clean (default-clean) @ cloud-scheduling-lab ---
[INFO] Deleting /home/vboxuser/CloudSchedulingLab/target
[INFO]
[INFO] --- resources:3.3.1:resources (default-resources) @ cloud-scheduling-lab ---
[INFO] skip non existing resourceDirectory /home/vboxuser/CloudSchedulingLab/src/main/resources
[INFO]
[INFO] --- compiler:3.13.0:compile (default-compile) @ cloud-scheduling-lab ---
[INFO] Recompiling the module because of changed source code.
[INFO] Compiling 1 source file with javac [debug release 17] to target/classes
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  0.838 s
[INFO] Finished at: 2026-11-10T05:08:43Z
[INFO] ------------------------------------------------------------------------
```

5.2 Run:

```
vboxuser@Mona:~/CloudSchedulingLab$ mvn exec:java -Dexec.mainClass="lab.SchedulingExperiment" -Dexec.classpathScope=compile
[INFO] Scanning for projects...
[INFO]
[INFO] ----------------------< lab:cloud-scheduling-lab >----------------------
[INFO] Building cloud-scheduling-lab 1.0-SNAPSHOT
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- exec:3.7.0:java (default-cli) @ cloud-scheduling-lab ---

================================================
A: VM SpaceShared + Cloudlet SpaceShared
================================================
INFO
================= Starting CloudSim Plus 8.5.1 =================

INFO  0.00: #Datacenter 1 is starting...
INFO  Broker 2 is starting...
INFO  Entities started.
INFO  0.00: DatacenterBrokerSimple2: List of 1 datacenters(s) received.
INFO  0.00: DatacenterBrokerSimple2: Trying to create Vm 0 inside #Datacenter 1
INFO  0.00: DatacenterBrokerSimple2: Trying to create Vm 1 inside #Datacenter 1
INFO  0.00: HostSimple: Vm 0 is booting up right away in Host 0/#DC 1, since no startup delay (boot time) was set.
INFO  0.00: VmAllocationPolicySimple: Vm 0 has been allocated to Host 0/#DC 1
INFO  0.00: HostSimple: Vm 1 is booting up right away in Host 0/#DC 1, since no startup delay (boot time) was set.
INFO  0.00: VmAllocationPolicySimple: Vm 1 has been allocated to Host 0/#DC 1
INFO  0.10: DatacenterBrokerSimple2: Sending Cloudlet 0 to Vm 0 inside Host 0/#DC 1.
INFO  0.10: DatacenterBrokerSimple2: Sending Cloudlet 1 to Vm 1 inside Host 0/#DC 1.
INFO  0.10: DatacenterBrokerSimple2: Sending Cloudlet 2 to Vm 0 inside Host 0/#DC 1.
INFO  0.10: DatacenterBrokerSimple2: Sending Cloudlet 3 to Vm 1 inside Host 0/#DC 1.
INFO  0.10: DatacenterBrokerSimple2: All 4 waiting Cloudlets submitted to some VM.
WARN  20.10: CloudletSchedulerSpaceShared: Cloudlet 2 requested 2048 MB of VmRam but no amount is available., which delays Cloudlet processing.
```

```
INFO  20.21: VmSimple: Requesting Vm 1 destruction on Host 0/#DC 1. Shutdown will finish immediately (since no Vm shutDownDelay was set).
INFO  20.21: VmSimple: Requesting Vm 0 destruction on Host 0/#DC 1. Shutdown will finish immediately (since no Vm shutDownDelay was set).
INFO  20.21: #Datacenter 1: Vm 1 destroyed on Host 0/#DC 1.
INFO  20.21: #Datacenter 1: Vm 0 destroyed on Host 0/#DC 1.
INFO  Simulation: No more future events

INFO  CloudInformationService0: Notify all CloudSim Plus entities to shutdown.

INFO
================= Simulation finished at time 20.21 =================

Cloudlet   VM         Status       Start        Finish
0          0          SUCCESS      0.10         20.10
2          0          SUCCESS      0.10         20.10
1          1          SUCCESS      0.10         20.10
3          1          SUCCESS      0.10         20.10
Completed Cloudlets: 4 / 4

================================================
B: VM SpaceShared + Cloudlet TimeShared
================================================
INFO
================= Starting CloudSim Plus 8.5.1 =================

INFO  0.00: #Datacenter 1 is starting...
INFO  Broker 2 is starting...
INFO  Entities started.
INFO  0.00: DatacenterBrokerSimple2: List of 1 datacenters(s) received.
INFO  0.00: DatacenterBrokerSimple2: Trying to create Vm 0 inside #Datacenter 1
INFO  0.00: DatacenterBrokerSimple2: Trying to create Vm 1 inside #Datacenter 1
```

```
INFO  20.10: DatacenterBrokerSimple2: Cloudlet 0 finished in Vm 0 and returned to broker.
INFO  20.10: DatacenterBrokerSimple2: Cloudlet 2 finished in Vm 0 and returned to broker.
INFO  20.10: DatacenterBrokerSimple2: Cloudlet 1 finished in Vm 1 and returned to broker.
INFO  20.10: DatacenterBrokerSimple2: Cloudlet 3 finished in Vm 1 and returned to broker.
INFO  20.21: Processing last events before simulation shutdown.
INFO  20.21: DatacenterBrokerSimple2 is shutting down...
INFO  20.21: VmSimple: Requesting Vm 1 destruction on Host 0/#DC 1. Shutdown will finish immediately (since no Vm shutDownDelay was set).
INFO  20.21: VmSimple: Requesting Vm 0 destruction on Host 0/#DC 1. Shutdown will finish immediately (since no Vm shutDownDelay was set).
INFO  20.21: #Datacenter 1: Vm 1 destroyed on Host 0/#DC 1.
INFO  20.21: #Datacenter 1: Vm 0 destroyed on Host 0/#DC 1.
INFO  Simulation: No more future events

INFO  CloudInformationService0: Notify all CloudSim Plus entities to shutdown.

INFO
================ Simulation finished at time 20.21 ================

Cloudlet   VM         Status       Start        Finish
0          0          SUCCESS      0.10         20.10
2          0          SUCCESS      0.10         20.10
1          1          SUCCESS      0.10         20.10
3          1          SUCCESS      0.10         20.10
Completed Cloudlets: 4 / 4
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  0.488 s
[INFO] Finished at: 2026-11-10T05:17:32Z
[INFO] ------------------------------------------------------------------------
```